DEEP LEARNING LABORATORY

EXPERIMENT 12

CNN ARCHITECTURE AND HYPERPARAMETER COMPARISON

Aim

To compare CNN configurations with different numbers of convolutional filters and learning rates, and evaluate their performance using test loss and test accuracy on the MNIST handwritten-digit dataset.

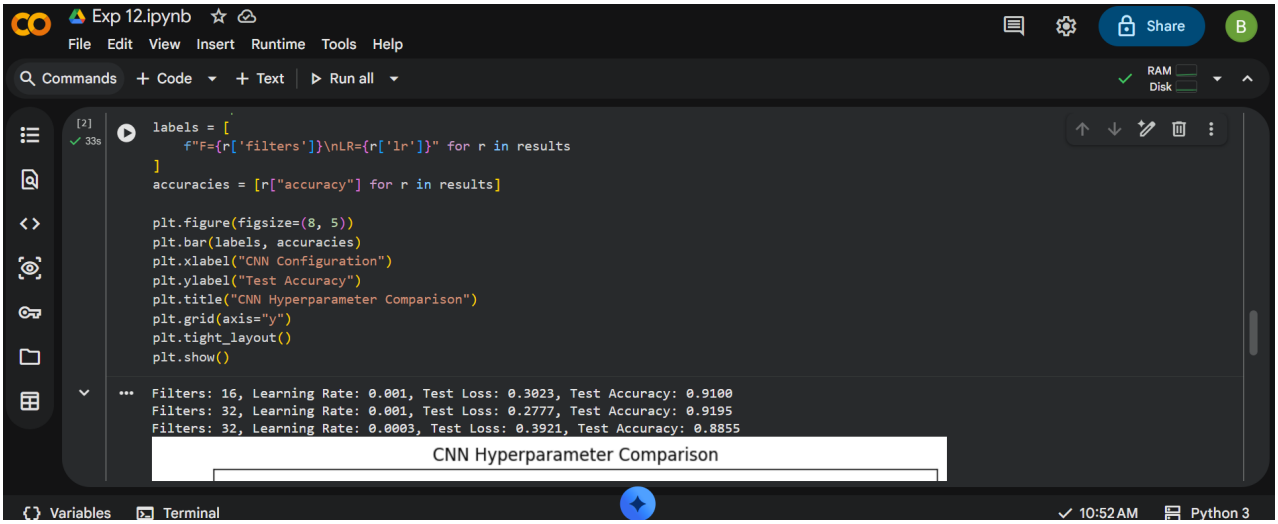
Observed Output

Three configurations were tested. The configuration with 32 filters and a learning rate of 0.001 achieved the highest test accuracy (91.95%) and the lowest test loss (0.2777) among the three configurations in this run.

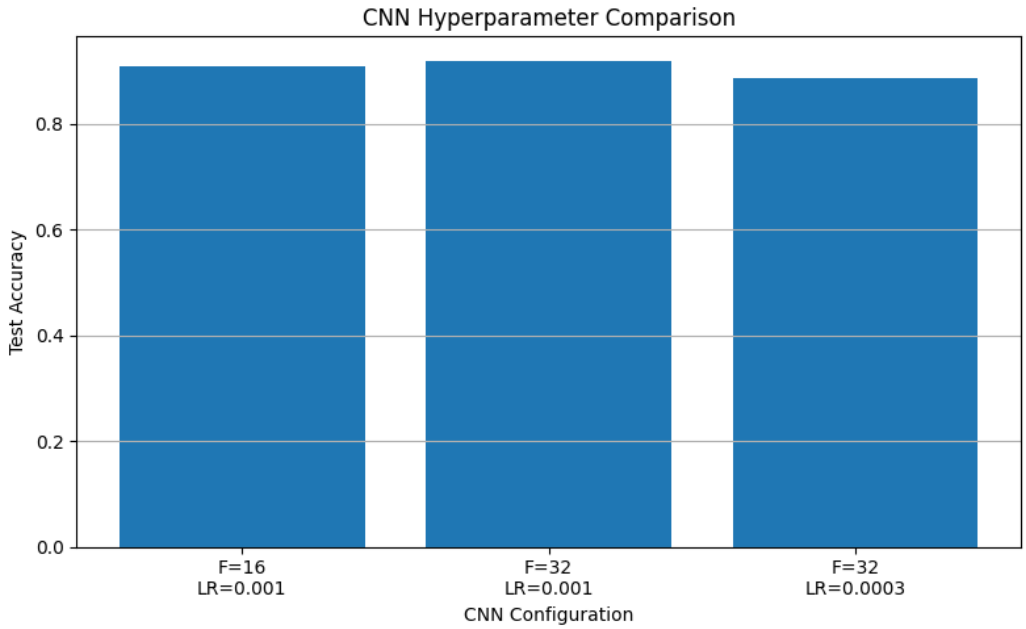
Result Table

CNN configuration	Test loss	Test accuracy
16 filters, LR = 0.001	0.3023	0.9100 (91.00%)
32 filters, LR = 0.001	0.2777	0.9195 (91.95%)
32 filters, LR = 0.0003	0.3921	0.8855 (88.55%)

Google Colab Output Screenshot



Test Accuracy Comparison Graph



Result

The experiment shows that CNN performance changes with filter count and learning rate. For this run, the 32-filter model with learning rate 0.001 performed best based on the recorded test accuracy and test loss. This conclusion applies to the tested configurations and dataset subset.